

# When does cortical coherence carry shared content?

## Compatibility and self-reconstruction as testable conditions

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20 September 2026

### Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that shared content and a self-model each need a condition beyond phase order, and state it. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on one simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies these implications; one

model satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a pre-registered decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

# 1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields, compared with perceptual content through independently fitted decoders (Section 10.2).

16 Binding-by-synchrony and communication-through-coherence accounts  
 17 treat phase relations as the medium of integration or of selective routing  
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the  
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both  
 20 leave open what fact about synchronized activity makes it one content rather  
 21 than coordinated activity carrying several. The contribution here is to sep-  
 22 arate the three requirements and prove which follow from which; each result  
 23 names a measurement that phase data alone cannot supply.

24 The objection that synchrony does not fix content is older than this work.  
 25 Critiques of the temporal binding hypothesis argued that synchrony, where  
 26 it occurs, cannot say which features belong together or what they jointly  
 27 represent: that grouping has to be computed before synchrony could sig-  
 28 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:  
 29 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality  
 30 and a pigeonhole count. What formalization adds to the informal argument  
 31 is quantitative and bookkeeping content. It gives the rate at which a co-  
 32 herence bound on content weakens with population size, the population at  
 33 which it stops constraining anything and how far chaining through overlap-  
 34 ping patches extends it, the diameter within which one contracting map can  
 35 reconstruct, and the number of codes a separated family of states requires.  
 36 It also gives a complete inventory of the premises a synchrony-to-content  
 37 argument consumes, each of which names a measurement, together with one  
 38 model in which all of them hold at once.

39 The familiar noisy mean-field oscillator supplies a stationary coherence  
40 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991); we add a minimum  
41 cost for changing coherence quickly. A Lean 4 development checks the im-  
42 plications, and a joint witness shows that the eight connecting hypotheses of  
43 the formal composition are jointly satisfiable.

44 The empirical question is which measurements could support or discon-  
45 firm the proposed connections. A phase resultant can miss a locally ordered  
46 winding field; stationary phase distributions can leave the coupling mecha-  
47 nism unidentified; and coherent adaptation can lose alignment with a desig-  
48 nated content template. We therefore distinguish stationary branches from  
49 finite observed onset, and compare prespecified field and connectivity onset  
50 models on held-out emergence recordings. The comparison tests a specified  
51 spatial prediction, not the presence of experience. Section 10.2 states the  
52 comparison that bears on conscious content directly.

53 The article makes six claims:

- 54 1. Compatible local descriptions determine a unique global state, while  
55 equal phases coexist with incompatible descriptions (Section 4).
- 56 2. Coherence bounds disagreement between overlapping descriptions  
57 through shared encoders of bounded sensitivity, by  $L\sqrt{2N(1-r^2)}$   
58 on the population read; on a simulated cortical sheet this constrains  
59 at most 10 sites, and chaining through overlapping patches extends it  
60 to 0.312 mm (Section 4.2).

- 61     3. A contracting self-map fixes one state and reconstructs within error  $\varepsilon$   
62         only a family of diameter at most  $2\varepsilon/(1 - \Lambda)$ ; reconstructing  $M$  states  
63         pairwise more than  $2\varepsilon$  apart takes  $M$  distinguishable codes (Section 5).
- 64     4. Changing coherence in time  $\tau$  has a minimum current cost that grows  
65         as  $1/\tau$  (Section 3.2).
- 66     5. Stationary phase statistics leave the coupling mechanism unidentified,  
67         finite ramps shift the observed onset, and a winding field keeps local  
68         order at global  $r = 0$  (Section 8).
- 69     6. The link to content predicts that, at matched phase coherence and  
70         single-region accuracy, decoders fitted independently on two regions  
71         agree more, beyond their separate accuracies, on perceived than on  
72         unperceived stimuli; equal excess counts against the link to content  
73         (Section 10.2).

74     Sections 2–5 develop the three requirements; Section 6 applies their mea-  
75     surement obligations to a digital candidate. Section 7 states what the formal  
76     composition establishes, and Sections 8–9 examine observables and the corti-  
77     cal test, and Section 11 states the scope of every result in one place. Table 1  
78     lists the connecting hypotheses the article argues from; Supplemental Mate-  
79     rial gives the full chain, derivations and controls. For a theoretical article the  
80     Methods are the definitions, proofs and simulations. Each result is stated in  
81     the section that uses it, Supplemental Material Table S1 maps it to its Lean  
82     declaration, and every simulation’s model, parameters and controls are given

83 in Supplemental Material. The Results are the theorems of Sections 3–5 and  
the numerical controls of Section 8.

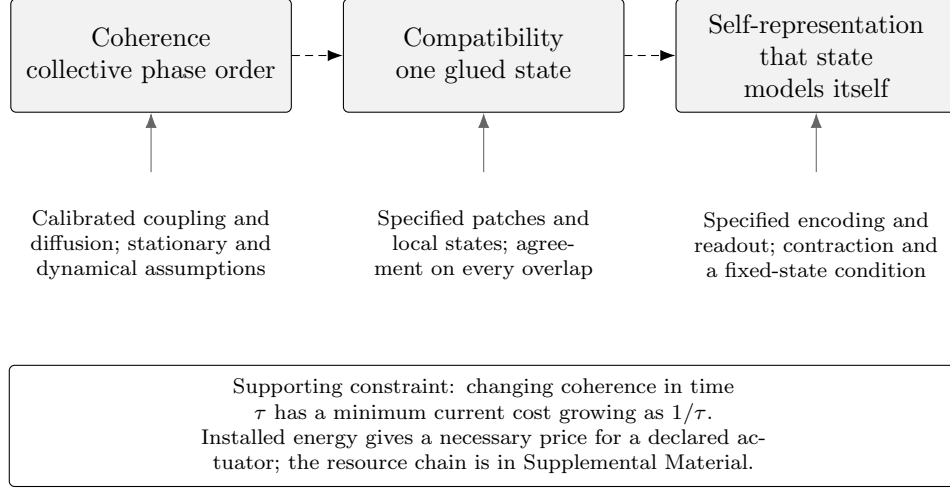


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections, each of which needs the inputs of the condition it points to in addition to phase coherence. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

84

## 85 2 Limits of phase order

86 Several limits hold without the cortical hypothesis. Equal phases can coexist  
87 with incompatible local descriptions; phase order constrains shared content  
88 only through encoders related on overlaps. A winding field can have zero  
89 global resultant while remaining locally ordered. A constant readout can have  
90 a unique fixed point without reconstructing its input. A region encodes only

the states it distinguishes, and by a declared deadline a site can report only changes inside its causal past. These limits reflect familiar phase oscillator and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one beside the condition or measurement it motivates.

### 3 Coherence: physical coupling and its cost

#### 3.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[ \omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here  $\theta_i$  is the phase of oscillator  $i$ ,  $\omega_i$  its natural frequency,  $D > 0$  is phase diffusion,  $W_i$  are independent Wiener processes, and  $K_{ij}$  has inverse-time units, with angles dimensionless. Uniform mean-field coupling is  $K_{ij} = K/N$ ; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized

107 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

108 where  $I_n$  are modified Bessel functions and  $\psi$  is the collective phase. The for-  
 109 mal derivation uses the absence of stationary current for a periodic gradient  
 110 drift to obtain this density, then verifies normalization and self-consistency.  
 111 At  $K \leq K_c = 2D$ , only  $r = 0$  is self-consistent (Sakaguchi, 1988; Strogatz &  
 112 Mirollo, 1991). Above threshold, exactly one positive solution coexists with  
 113 zero and increases strictly with  $K$ . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

114 Figure 2 shows this stationary branch.

115 Linearization about the uniform density gives rate  $K/2 - D$  for the first  
 116 harmonic and  $-Dn^2$  for harmonics  $n \geq 2$ . A growing mode therefore exists  
 117 exactly above threshold. About the coherent density, the quadratic form  
 118 is strictly dissipative transverse to collective rotation, whose rate is zero.  
 119 These linear estimates concern the limiting equation. Relating them to finite  
 120 stochastic trajectories requires a dynamical mean-field limit and branch selec-  
 121 tion; exchange symmetry alone does not establish the needed independence  
 122 (Sznitman, 1991).

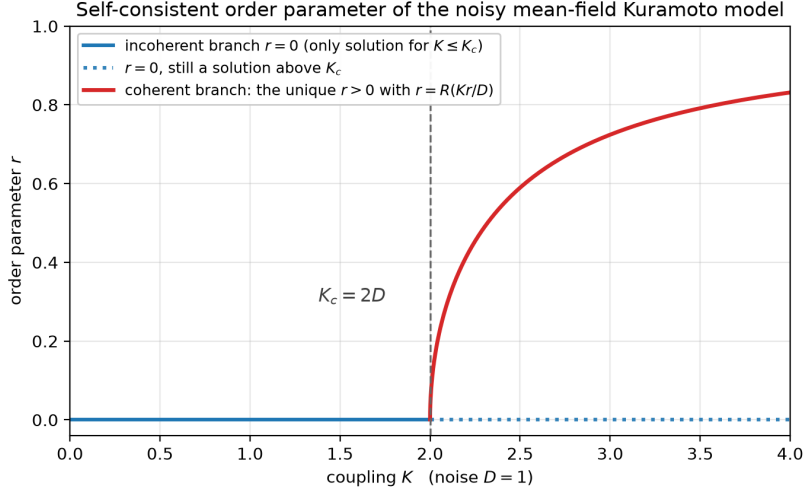


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion  $D = 1$ . The positive coherent branch appears continuously above  $K_c = 2D$  and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

### 123 3.2 A speed limit on changes of coherence

124 One thermodynamic statement bears on coherence directly. For a smooth  
 125 positive periodic phase density with probability current  $J$  and diffusion  $D$ ,  
 126 the current functional  $\sigma_J = \int J^2 / (D\rho) d\theta$  satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

127 while  $0 < r < 1$ . A change of coherence completed in time  $\tau$  therefore has a  
 128 minimum current cost that grows as  $1/\tau$ , whereas holding a stationary order  
 129 has no positive floor. Read the other way, a current budget  $C$  sets a minimum

130 duration  $\tau \geq [\Delta \arcsin r]^2 / (DC)$  for a given change. This bears on ignition-  
 131 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,  
 132 2014), and on recovery latency after anaesthesia: each has a floor set by the  
 133 current available to drive it. The bound is a heat or power statement only  
 134 when  $J$  is identified with a physical flux, a reservoir at a stated temperature  
 135 is specified and local detailed balance holds for the steps concerned; only  
 136 then does  $\sigma_J$  convert to entropy production (Landauer, 1961; Norton, 2005;  
 137 Shenker, 2000). Installed energy supplies a separate, necessary price: for  
 138 a declared actuator whose response modes carry installation prices, mean  
 139 coupling is at most  $\kappa U_{\text{inst}}$  for a mode-data factor  $\kappa$ , so a coherent stationary  
 140 branch requires installed energy  $U_{\text{inst}} > 2D/\kappa$ . Supplemental Material gives  
 141 both derivations, the register and feedback budgets, and the sensitivity of  
 142 the metabolic comparison to its assumptions.

## 143 **4 Compatibility: from coordinated activity** 144 **to a global description**

### 145 **4.1 Why common timing is insufficient**

146 The formal setting is a sheaf: each patch of the substrate carries local de-  
 147 scriptions, and restriction maps cut a description on a patch down to any  
 148 smaller patch. A *section* is a description on one patch, and a *global section* is  
 149 a description on the whole substrate. For overlapping patches  $U$  and  $V$ , com-

150 patibility means that restricting either local description to  $U \cap V$  gives the  
151 same object. Descriptions can agree on shared quantities while contributing  
152 different information elsewhere. Compatible sections of the specified sheaf  
153 have a unique global extension. On a finite cover, local finite spatial mea-  
154 sures agreeing as measures on every overlap also have a unique finite-measure  
155 extension whose sheaf image is the glued section. No mass-one normalization  
156 is imposed.

157 In the cup example, visual and reaching descriptions must express a  
158 shared quantity, such as cup location, in common coordinates. Restriction  
159 retains that quantity while dropping details particular to each description.  
160 Agreement permits surface appearance and hand configuration to contribute  
161 to one situation. The mathematical counterexample assigns the same phase  
162 to every patch while two local mass profiles disagree on a shared site. Those  
163 profiles have no common extension. Because they are chosen independently  
164 of phase, synchrony places no constraint on their disagreement.

165 This distinction limits the biological interpretation. Receptive territories  
166 and content domains must be related to spatial patches, and the descriptions  
167 must support the required extension. Matching overlap marginals of de-  
168 coded probability laws need not determine a unique global joint distribution  
169 (Abramsky & Brandenburger, 2011).

170 The cover is itself a choice, and the construction constrains it in one way  
171 only: synchronized patches agree where they overlap. Where they do not  
172 overlap there is nothing to compare, so a cover by pairwise disjoint patches

173 satisfies that obligation for *every* assignment of local descriptions and phases  
174 and the gluing theorem still returns a unique global section, which is the  
175 tuple it was handed. Membership of the class then reports nothing about the  
176 system beyond the patches covering it. What a gluing claim asks of a cover  
177 is therefore a non-degeneracy the theorem does not assume: that distinct  
178 patches meet, and that what they share is a quantity fixed independently of  
179 the agreement it is being used to establish.

180     The cup example exposes a second gap, between two kinds of overlap.  
181 The sheaf is built over a spatial substrate, so its patches overlap where they  
182 share sites. Visual and motor territories that agree on cup location overlap  
183 in what they describe, not in where they lie, and the construction supplies  
184 no relation between the two. One route recasts the construction over content  
185 variables: each region becomes a partial decoder whose domain is the set of  
186 variables it estimates, and two regions overlap exactly where they estimate  
187 a common variable. Gluing and approximate selection hold on such a cover,  
188 and there compatibility is a consequence: regions decoding a variable within a  
189 tolerance agree on it within twice that tolerance, so only agreement beyond  
190 fidelity is evidence. The phase-to-content bounds concern spatial patches  
191 and do not transfer. One principle selects such a cover without presupposing  
192 the agreement it is used to test. Assign each region the variables that can  
193 be decoded from its activity above a declared criterion, on data held out  
194 from the agreement test, and let two regions overlap on exactly the variables  
195 decodable from both. Decodability is fixed by what each region carries before

any comparison is made, so the cover cannot be chosen to make agreement come out. It is non-degenerate exactly when its nerve is connected, meaning that every region is linked to every other through a chain of shared decodable variables, and recordings can fail that condition. The principle has three inputs, the candidate variables, the criterion and the decoder class, and each has a rule that does not consult agreement.

The candidate variables are those the experiment manipulates and the task makes behaviourally relevant, fixed before recording. Data-driven alternatives, such as directions of maximal shared variance between two regions, select for the agreement under test and are excluded. The criterion becomes a statistical one: a region encodes a variable when its cross-validated decoding exceeds a permutation null at a pre-registered level, and results are reported across a declared range of levels. The level need not be exact, because two accurate decoders agree automatically (Eq. (13)); the statistic of Section 10.2 is agreement in excess of what each region’s accuracy implies, which remains valid at any admitted level. The decoder class is bounded on both sides. Unrestricted nonlinear decoders recover a variable from any region carrying information about it, so nearly every pair of regions overlaps and the cover again constrains nothing. The positive bound of Section 4.2 requires encoders with a declared Lipschitz constant  $L$ , which enters  $\delta_r$  directly, so the class should consist of  $L$ -Lipschitz maps; linear readout, the usual operationalization of information explicitly available to downstream neurons (DiCarlo & Cox, 2007; Kriegeskorte & Diedrichsen, 2019), is one such class. Within the

219 admitted family, the decoder and the distance in which content discrepancy is  
 220 measured are chosen by held-out decoding accuracy and never by agreement.  
 221 Neither is neutral: which similarity measure best describes neural activity  
 222 differs across tasks (Bobadilla-Suarez et al., 2020). What remains is that the  
 223 cover is the experimenter’s. Whether cortex realizes unity over the chosen  
 224 variables is the empirical claim under test, and no choice of decoder settles  
 225 it. The correlate is therefore operational relative to these pre-registered in-  
 226 puts, and carrying the formal results to such a cover is the recasting sketched  
 227 above.

## 228 **4.2 When coherence does constrain content**

229 A declared relation between phase and content gives a positive result. Let  
 230 contents take values in a real normed vector space  $\mathcal{C}$ . Patch  $i$  reads an  
 231 assigned oscillator through  $e_{i,x}$  at site  $x$ , evaluated on the circle point  
 232  $(\cos \theta, \sin \theta)$ . Assume a uniform Lipschitz constant  $L$  in chord distance,  
 233 meaning that an encoder’s output moves by at most  $L$  times the chord  
 234 between two input phases, and require  $e_{i,x} = e_{j,x}$  wherever patches  $i$  and  
 235  $j$  overlap. Encoders may vary across sites; their equality on overlaps is a  
 236 supplied relation between descriptions of the shared quantity. Write  $z_k = e^{i\theta_k}$   
 237 and  $m = N^{-1} \sum_k z_k$ , so  $|m| = r$ . The phasors’ squared deviations from their  
 238 mean sum to  $\sum_k |z_k - m|^2 = N(1 - r^2)$ , and  $|z_i - z_j| \leq |z_i - m| + |z_j - m|$   
 239 gives  $|z_i - z_j|^2 \leq 2N(1 - r^2)$  for each pair. Consequently every overlap

240 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (5)$$

241 Here discrepancy is the norm of the difference, and  $r$  and  $N$  are read on  
242 whichever population the bound is applied to. That choice is what decides  
243 whether the estimate says anything. A patch carries an order parameter  
244 of its own, and reading the bound there carries that patch's resultant and  
245 that patch's site count, so a large population observed through small patches  
246 is constrained by the small count. Read instead on the whole array, the  
247 factor  $\sqrt{N}$  makes the estimate weak, because it draws a pointwise guarantee  
248 from an average over  $N$  sites. The order cannot be improved in general:  
249 one oscillator in antiphase among  $N$  leaves  $N(1-r^2)$  near 4. At  $r=1$  the  
250 observations agree exactly at either scale. A shared scalar encoder is a special  
251 case. Dropping the overlap relation permits disagreement even at perfect  
252 locking. A common nonexpansive linear projection preserves agreement.

253 What Eq. (5) bounds is  $L$  times a chord between phasors, and a chord  
254 never exceeds 2, so the estimate constrains anything only where  $\sqrt{2N(1-r^2)}$   
255 falls below 2, whatever the value of  $L$ . On the simulated sheet of Section 8,  
256 at its patch-local order for a coupling decay length of 0.1 mm, the bound is  
257 informative for at most 10 sites; this count and the chain's reach below are  
258 values for that sheet and move with the decay length and patch order. Three  
259 conditional results decline the step that puts  $\sqrt{N}$  there. A pair count bounds

260 the fraction of disagreeing pairs at any population size but names no pair.  
 261 A chain across the sites that overlapping patches share compares distant  
 262 territories through the patches between them, with error growing linearly in  
 263 hops rather than with population: on the same sheet it stays informative  
 264 for 19 hops, 0.312 mm. A spectral gap removes the population count at a  
 265 frequency-locked configuration, given a declared algebraic connectivity of the  
 266 coupling graph. Territories whose agreement unity concerns lie centimetres  
 267 apart, beyond the chain's reach, and across them the coherence-to-content  
 268 link is the bridge assumption E78 (Table 1). Supplemental Material gives  
 269 the three derivations and the sheet's numbers.

270 Mix retained fields with observations at gain  $\eta \in [0, 1]$ :  $s_i \mapsto (1 - \eta)s_i +$   
 271  $\eta o_i$ . Content and observation discrepancies bounded by  $\varepsilon$  and  $\delta$  give a next  
 272 discrepancy at most  $(1 - \eta)\varepsilon + \eta\delta$ . Repeated observations bounded by  $\delta$   
 273 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

274 Thus agreeing observations preserve exact compatibility and, for  $\eta > 0$ , re-  
 275 duce an initial mismatch. Approximate observations leave a nondecaying  
 276 term in the error bound. Taking a uniform bound on Eq. (5) connects this  
 277 update to the phase process.

### 278 4.3 What approximate agreement determines

279 Exact gluing requires exact restrictions. For nonnegative profiles on a finite  
280 substrate, suppose every overlap discrepancy is at most  $\varepsilon$  in uniform distance.  
281 Weights that vanish outside their patches and sum to one at each site select  
282 a global profile by weighted averaging. The selection lies within  $\varepsilon$  of each  
283 patch there, and any two such selections differ by at most  $\varepsilon$ , independently  
284 of cover multiplicity. An  $L$ -Lipschitz readout consequently varies by at most  
285  $L\varepsilon$  across selections.

286 At zero tolerance, the selection is the unique exact extension. At every  
287 positive tolerance there are families with no exact extension; selection then  
288 gives an approximation whose dependence on weights is bounded. Supple-  
289 mental Material gives a three-site witness attaining the bound. The mass-  
290 profile construction uses the scalar case of the content dynamics and addi-  
291 tionally requires nonnegativity of the initial profiles and observations.

292 These constructions concern declared objects: a specified cover, contents  
293 valued in a normed space, and encoders related on overlaps. Section 8 sets  
294 out what a measurement of their neural counterparts would have to establish.

## 295 5 Self-representation: a state that models it- 296 self

### 297 5.1 The reflexive construction

298 Compatibility gives a global state. Reflexivity relates it to an encoding within  
299 that state. Equip global sections with a complete metric. *Restriction reso-*  
300 *nance* is the additional condition that the encoding is the state’s own restric-  
301 tion to a region  $\mathcal{A}$ , so that the region encodes the state by being part of it.  
302 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

303 The region and readout are fixed across the state space. If  $F$  contracts and  
304 fixes the section  $s_*$  formed from the local descriptions, Banach’s theorem  
305 makes  $s_*$  its unique fixed point (Banach, 1922). The formal bridge E89  
306 (Table 1 gives the full composition and its edges) supplies the contraction  
307 and  $F(s_*) = s_*$ ; restriction resonance is separate. Supercritical coupling  
308 makes the stipulated contraction factor  $\Lambda = e^{-(K-2D)\tau/2}$  less than one for a  
309 relaxation time  $\tau > 0$ ; Section 11 states the standing of that factor. A fixed  
310 contraction can reconstruct accurately only a restricted family of states: if  
311 its error is at most  $\varepsilon$  throughout that family, any two members are at most  
312  $2\varepsilon/(1 - \Lambda)$  apart.

313 In the cup example, the global state includes the represented scene and the

314 perceiver's relation to it. A local encoding must retain enough information  
 315 for the readout to recover that modelled state; cup position alone would be  
 316 insufficient if hand configuration and viewpoint could not be reconstructed.

317 For the same map with Lipschitz factor  $\Lambda < 1$ , approximate reconstruc-  
 318 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

319 At the stipulated rate this radius grows as  $2\varepsilon/((K - 2D)\tau)$  near threshold, for  
 320 fixed positive  $\varepsilon$  and  $\tau$ . Exact uniqueness survives while the error guarantee  
 321 weakens.

## 322 5.2 Reconstruction across a family of states

323 Sensitivity can be required quantitatively. Declare relevant macrostates, an  
 324 encoder  $\Phi$  and readout  $R$ , and require  $d(s, R(\Phi(s))) \leq \varepsilon$  throughout that  
 325 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

326 Thus  $M$  relevant states pairwise more than  $2\varepsilon$  apart require at least  $M$   
 327 distinguishable codes; an alphabet of at most  $2^b$  codes requires  $M \leq 2^b$ .  
 328 Applied to the reflexive boundary,  $E$  is its local encoding and  $R \circ E$  is the  
 329 existing self-prediction map.

330 This is an additional accuracy criterion on a family, with no requirement  
 331 for multiple exact fixed points. For the same fixed contracting map used by

332 E89, however, the diameter bound above is an additional restriction: at zero  
 333 error the family contains at most one metric-distinct state. A broad family  
 334 needs an input-conditioned map, one contraction  $F_u$  per input  $u$ . If  $F$  is  
 335  $L$ -Lipschitz in  $u$ , members with inputs  $u, v$  lie within  $(2\varepsilon + L d(u, v))/(1 - \Lambda)$   
 336 and the fixed points within  $L d(u, v)/(1 - \Lambda)$ , so a self-state can follow the  
 337 scene. The code criterion then binds within each input: a readout storing  
 338 one state per input reconstructs without encoding anything. Under the fixed  
 339 map of E89 the minimal-self correlate is a condition on one self-state. A blind  
 340 encoder or constant readout fails the criterion for separated states despite  
 341 having a unique fixed point.

342 Where the encoding is read as the state's own restriction to a region, the  
 343 criterion becomes a condition on the region. Restriction resonance requires  
 344 the encoder and that restriction to identify the same pairs of states, so a  
 345 boundary reconstructing two relevant states more than  $2\varepsilon$  apart while its  
 346 region reads them identically is not resonant. Read forwards, a resonant  
 347 boundary that reconstructs a pairwise separated family restricts injectively  
 348 on it. Which states a region reads identically is fixed by where the region  
 349 sits, before an encoding is chosen, so the resolution the family demands is  
 350 the region's and no readout supplies it. The implication runs one way: a  
 351 region that does resolve the family is not thereby shown to carry a resonant  
 352 encoding.

353 Resonance and resolution are jointly satisfiable without a local fold. Take  
 354 the region to be the whole substrate and use the identity readout: restric-

tion is the identity, so this boundary is resonant and reconstructs every de-  
 clared family exactly. Its map has Lipschitz factor one and does not satisfy  
 E89’s strict contraction premise on a nontrivial metric space. A properness  
 condition on the region is therefore a separate requirement, and the fixed-  
 contraction diameter bound remains in force when E89 is also required. With  
 an  $L$ -Lipschitz readout, a family reconstructed within  $\varepsilon$  whose members lie  
 pairwise more than  $d_{\min}$  apart has  $\delta$ -separated codes whenever  $\delta L \leq d_{\min} - 2\varepsilon$ ,  
 so the family is no larger than the  $\delta$ -packing number of what the region car-  
 ries. At the whole substrate it says nothing, because the region’s range is  
 everything. The gain  $L$  is a declared input, and the bound weakens without  
 limit as the gain grows.

### 5.3 Agreement after a change takes communication

Snapshot compatibility does not specify how quickly descriptions respond  
 when a shared quantity changes. In a synchronous network whose updates  
 read only the site’s own state and declared neighbours’ messages, two initial  
 configurations agreeing on a site’s causal past through  $T$  rounds produce the  
 same state there at round  $T$ . The deadline consequence is the one Section 2  
 states, and it follows from the reconstruction bound in Section 5.2: a report  
 from that site is a readout of a code that the two interventions leave identical.

A three-site delay line witnesses failure after one round and exact report-  
 ing after two; cutting the path prevents delivery at every round. A report  
 may still happen to be correct about one fixed world. The bound concerns

377 responsiveness across interventions.

378     Throughout this section *prediction* names a reconstruction map, and the  
379 relevant family of states, the metric, the tolerance, the communication graph  
380 and the deadline are all declared inputs, as are the represented variables and  
381 the fidelity counted as accurate. Reading the map as temporal neural pre-  
382 diction would require dynamics; reading the resulting code count as physical  
383 memory capacity would require a hardware model, which a state-dependent  
384 readout would have to extend.

## 385   **6   What the conditions ask of a digital candi-** 386       **date**

387 A claim that a digital system, such as a large language model (LLM) run-  
388 ning on graphics processing units (GPUs), has unified content or represents  
389 itself commits its maker to the measurements this account asks of cortex.  
390 The maker must identify the shared quantities the system’s parts describe,  
391 show that decoders fitted independently on those parts agree about them,  
392 and show that a declared local code separates the relevant family of states  
393 and survives intervention. The indicator-properties approach assesses arti-  
394 ficial systems against computational properties drawn from theories of con-  
395 sciousness (Butlin et al., 2023); the conditions here are fewer and each is  
396 a measurement rather than an architectural property. A digital system is  
397 also where they can be measured today, with full access to its state and

398 exact interventions on it: fitting decoders for one shared quantity on two  
399 separate layers, heads or token positions, and testing their agreement and  
400 whether reconstruction survives intervention, would carry out the proposed  
401 measurement directly.

402 One constraint follows from the architecture alone. Within one forward  
403 pass of a causally masked model, position  $t$  reads only its causal past, so an  
404 intervention outside that past cannot be reconstructed at  $t$  by that step. This  
405 is the deadline result of Section 5 applied to the mask, and it bounds which  
406 variables a position can encode about the rest of the sequence within one  
407 step. A second constraint needs a premise the architecture does not supply:  
408 a linear encoder and readout factoring through  $d$  coordinates pay reconstruc-  
409 tion error at least the discarded spectral tail of the coupling operator, but  
410 query/key dimension does not supply such a bottleneck for softmax atten-  
411 tion. These results rank no architecture family, imply no general inferiority of  
412 GPU hardware and settle no question of machine experience. Supplemental  
413 Material gives the bottleneck derivation, a synthetic two-view task with its  
414 controls, the information ceiling of token outputs and the hardware scope.

## 415 **7 The formal composition**

416 The Lean development composes the three requirements into one conditional  
417 theorem. Its passive and feedback branches each accept eight connecting hy-  
418 potheses, running from finite capacity through a register, dissipation and a

419 continuum coupling to the field, coherence, the cover and the self-prediction  
 420 fixed point. Each edge is a hypothesis of the composition, and its inferen-  
 421 tial content is its last step, where contraction and a fixed glued section give  
 422 uniqueness. Table 1 lists the four edges the article argues from; Supplemen-  
 423 tal Material gives the full chain, its thermodynamic edges and the register  
 424 ledger of the feedback branch. Both branches also take a cover carrying ex-  
 425 act overlap compatibility and a reflexive boundary specifying encoding and  
 426 readout, and conclude that the glued section is the unique fixed point of self-  
 427 prediction. Composing the edges establishes that the premises are consistent,  
 428 named in one place and machine-checked together, and joint witnesses satisfy  
 429 every edge of each branch at once. The development declares no additional  
 430 axioms beyond its libraries’ foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

| Connection                             | Status                 | Required assumption                                                                            |
|----------------------------------------|------------------------|------------------------------------------------------------------------------------------------|
| E56<br><i>field<br/>identification</i> | Physical<br>commitment | A specified physical field realizes the limiting coupling and satisfies the kernel conditions. |
| E67<br><i>above<br/>threshold</i>      | Physical<br>commitment | That field operates above the synchronization threshold.                                       |

Table 1 (continued)

| Connection                               | Status               | Required assumption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E78<br><i>coherence to cover</i>         | Bridge assumption    | Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity $\lambda_2$ instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of $\kappa$ . |
| E89<br><i>contraction to fixed point</i> | Modelling assumption | A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$ . Reconstruction over a declared family imposes an additional code-separation criterion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## 431 8 What the observations establish

432 The composition holds under its premises. Whether observation can estab-  
433 lish those premises is a separate question, and the three requirements it  
434 connects concern different objects: phase dynamics, local descriptions and  
435 internal encodings. Their observables must retain these distinctions. The

436 following controls show which mechanisms remain indistinguishable, which  
 437 temporal approximations fail, which states one observable cannot separate,  
 438 and what independent content measurements must establish. Each is a finite  
 439 numerical regime on a declared model, with full parameters and controls in  
 440 Supplemental Material, and none of them calibrates a cortical quantity.

## 441 **8.1 Stationary distributions leave the mechanism un-** 442 **determined**

443 Equation (2) gives a concentration–coherence relation with no fitted shape  
 444 parameter. Estimate  $r$  from circular resultant length and  $a$  from density  
 445 shape: inverting  $R$  from the measured  $r$  would make agreement automatic.  
 446 Even independently estimated agreement identifies a statistical family shared  
 447 by different mechanisms.

448 Let  $q(t)$  cross 2, let  $r(q)$  be its stationary branch, and fix  $D_0 > 0$ . Increas-  
 449 ing coupling  $K(t) = D_0 q(t)$  at diffusion  $D_0$  and decreasing diffusion  $D(t) =$   
 450  $K_0/q(t)$  at coupling  $K_0$  both give concentration  $a = q(t)r(q(t))$ . Uncoupled  
 451 oscillators with prescribed drift  $-h(t)\sin(\theta - \psi)$ , where  $h(t) = D_0 q(t)r(q(t))$ ,  
 452 and diffusion  $D_0$  have that same density. Their stationary distributions co-  
 453 incide window by window despite different causes.

454 Resolved dynamics supply additional information. In the ideal model  
 455 with continuous latent paths, quadratic variation accumulates at rate  $2D$   
 456 and conditional increment means constrain drift; a filtered observation of

457 the phase needs its own observation model before either is read off. Separat-  
 458 ing endogenous coupling from a matched common drive additionally requires  
 459 a measured drive or controlled perturbation; identifying electromagnetic cou-  
 460 pling requires evidence specific to that interaction. Supplemental Material  
 461 gives the equivalence construction and its numerical check. Phase statistics  
 462 pooled over time and over bipolar pairs cannot supply these estimates: on  
 463 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-  
 464 fectly phase-locked signal at  $r = 0.0001$ , so a spatial test needs an estimand  
 465 that removes the common rotation and a montage whose effect on spatial  
 466 phase is modelled; Supplemental Material reports this estimator exercise and  
 467 the dependence-aware design.

## 468 **8.2 Finite ramps change the observable onset**

469 The square-root law in Eq. (3) describes stationary solutions, and a changing  
 470 coupling needs time for the phase distribution to approach them. In a deter-  
 471 ministic mean-field control crossing the threshold, none of the scanned rates  
 472 meets the declared tracking tolerance at onset. In a finite-population ramp  
 473 study ( $N = 2000$ ,  $D = 1$ , 32 replicas, 8 speeds), faster ramps delay escape.  
 474 The fitted delay exponent 0.444 has a replica-bootstrap interval  $[0.398, 0.496]$   
 475 that excludes the ideal  $1/2$ , a deterministic reference shows that the escape  
 476 criterion changes the fitted exponent, and no common asymptotic exponent  
 477 is established. Delayed passage through a noisy bifurcation (Berglund &  
 478 Gentz, 2002) makes the apparent exponent depend on ramp speed and fit-

479 ting window. Supplemental Material gives the sweep and its figure, the  
 480 fluctuating-coupling and population-size controls, and the signed-network  
 481 crossing bracket.

### 482 **8.3 A winding state leaves the order parameter am-** 483 **biguous**

484 The global order parameter  $r$  discards spatial arrangement. On a periodic  
 485 sheet, a phase field can wind once around the sheet and have  $r = 0$  while  
 486 neighbouring sites remain locked. At identical frequencies, every such wind-  
 487 ing is stationary for an isotropic symmetric kernel: contributions at opposite  
 488 separations cancel. It violates the quarter-turn initial-spread premise of the  
 489 convergence result, which is why that result can still select the uniform state.

490 In the spatial control, the winding persists below the interpolated de-  
 491 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform  
 492 and winding starts under the same coupling, diffusion and kernel end at  
 493  $r = 0.9657$  and  $r = 0.0083$ ; patch-local order is 0.9015. Under quenched fre-  
 494 quency spread  $1.5 \text{ rad s}^{-1}$ , the measured retention boundary is 0.1421 mm,  
 495 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen  
 496 realizations. These are finite controls, not a cortical wave identification or a  
 497 derived transition.

498 Linearization about a winding gives explicit mode rates, which concern  
 499 linear stability rather than the measured retention boundary. An amplitude

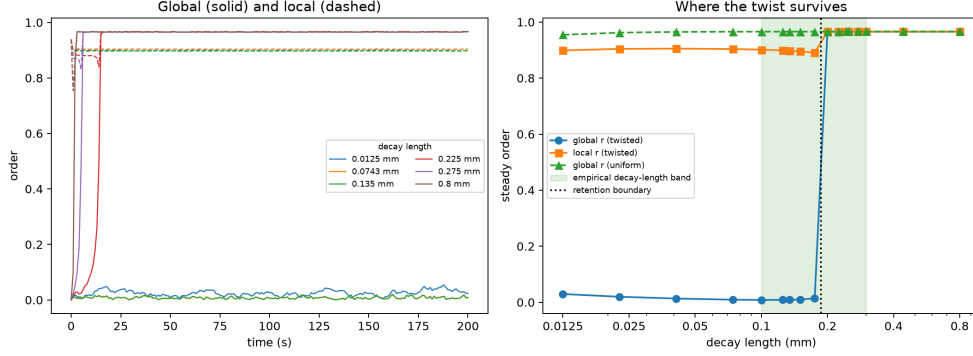


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of  $r$  at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked.

500 extension adds an existence condition under which a broad kernel can remove  
 501 a winding that the phase-only equations retain as a stationary state.

502 Neither model turns these states into measured cortical spirals. The pe-  
 503 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has  
 504 no propagation speed, and the amplitude variable has no calibrated relation  
 505 to a recorded envelope. Supplemental Material gives the spectral, amplitude  
 506 and defect results. Locally, Eq. (5) can still bound content disagreement on a  
 507 winding patch when shared encoders are supplied, although its global  $r = 0$   
 508 form is uninformative. A nonzero ring winding number  $q$  also obstructs a  
 509 global real-valued phase lift; it has no identified content meaning here. Thus  
 510 an onset criterion based only on  $r$  can register a change of basin as well

511 as a change of coupling. The protocol of Section 9.3 needs a spatial phase  
512 observable.

## 513 **8.4 Coherent adaptation can lose environmental align-** 514 **ment**

515 An adaptive coupling rule might supply representational structure, but its  
516 objective must reward the relevant agreement. For symmetric coupling, pair-  
517 wise sine terms cancel in the population mean drift. Minimizing squared  
518 drift at that fixed mean therefore rewards equalizing drift across nodes. In  
519 the frequency-cluster control, this favors coupling between unlike-frequency  
520 clusters, whereas the diagnostic environmental template rewards coupling  
521 within clusters. Supplemental Material proves descent of this objective for  
522 a specified gradient flow at fixed phases and distinguishes it from entropy  
523 production.

524 The finite control co-evolves noisy phases and symmetric, nonnegative  
525 couplings at fixed total resource. Its template is a diagnostic covariance  
526 of latent signals, never a training input. Across three illustrative seeds,  
527 second-half order remains between 0.9167 and 0.9222, while the within-  
528 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–  
529 0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel con-  
530 trols are reported in Supplemental Material.

531 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

## 8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at  $AUC = 0.986$ , on a 95% interval of  $[0.980, 0.992]$  resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

On constructed content, phase carries a share 1.000 of the content entropy

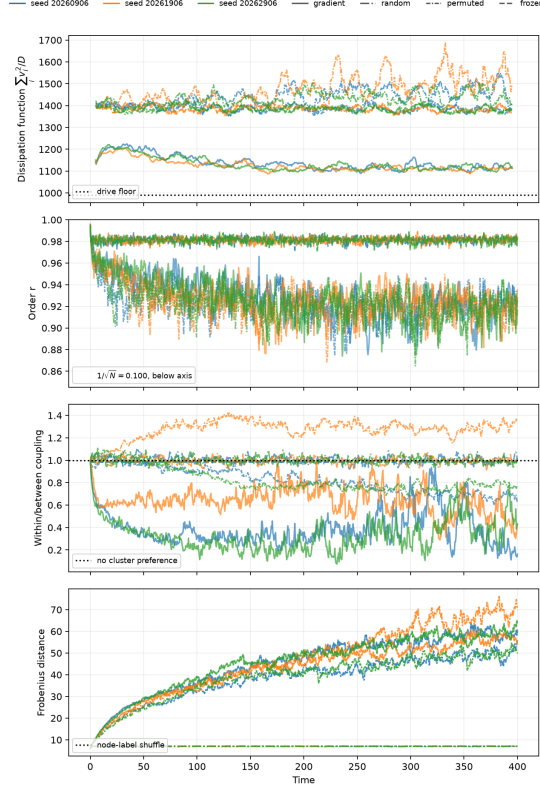


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. Section 4 gives the decodability rule that fixes the cover without consulting agreement.

## 9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

### 9.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models

573 propose coordination of distributed neural information (Pinotsis & Miller,  
574 2022; McFadden, 2020). Phase organization under propofol motivates study-  
575 ing this regime (Bastos et al., 2021).

576 The formal field predicate requires a jointly continuous kernel, a normal-  
577 ized substrate, coupling equal to the kernel average, and no single site pro-  
578 viding more than half that strength. Toy constructions satisfy it. A cortical  
579 realization must specify the scale at which the kernel and its reduction ap-  
580 ply: calcium imaging reports different state-dependent modularity patterns  
581 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

582 Cortical columns are a serious candidate for the oscillator index. They are  
583 delineated by response properties and anatomy rather than by the agreement  
584 they would be used to establish (Mountcastle, 1997), and a phase can be  
585 estimated from a column-sized territory at the electrode spacings used for  
586 cortical wave measurements (Townsend et al., 2015). They are a poor content  
587 cover: non-overlapping columns form a disjoint cover (Section 4), much of  
588 association cortex has no accepted columnar parcellation, and the column’s  
589 standing as a canonical unit is contested (Horton & Adams, 2005; Rakic,  
590 2008). The content cover is the decodability cover of Section 4, and the  
591 bridge assumption E78 records its identification with the oscillator index.

592 The spatial reduction is also testable. A row-sum aggregation at zero  
593 diagonal tracks the uniform-coupling control over the tested spatial decay  
594 range, while frequency heterogeneity displaces the operational threshold. The  
595 fit window biases both estimates relative to  $2D$ ; agreement with the control

596 is therefore the supported conclusion. Supplemental Material reports the  
 597 reduction, its numerical error and the distinct assumptions needed to identify  
 598 the kernel average with the scalar coupling of the installed-energy condition.

599 Converting a spike-time shift into coupling needs a rate: the dimensionless  
 600 product of population size  $N$ , timing sensitivity  $\mathcal{T}$ , field amplitude  $E$  and  
 601 frequency  $f$  becomes a coupling through a conversion rate  $\gamma$ ,

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

602 This rate describes how induced timing shifts accumulate into drift.

## 603 9.2 Extracellular geometry and recovery

604 The proposed control variable is extracellular geometry. Extracellular vol-  
 605 ume fraction is approximately  $\alpha = 0.2$  at tortuosity  $\lambda \approx 1.6$  (Syková &  
 606 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie  
 607 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively  
 608 at  $0.3\text{--}0.6 \text{ S m}^{-1}$  (Barbour, 2017). At fixed transmembrane current density,  
 609 an effective-medium description predicts that a larger extracellular fraction  
 610 raises conductivity and reduces field amplitude.

611 The coupling-gated hypothesis consequently predicts that extracellular  
 612 contraction on awakening raises  $E$  and, holding the other factors in Eq. (10)  
 613 fixed, raises effective coupling. This argument supplies a direction; the  
 614 geometry–coupling relation, its magnitude and its role in human recovery

615 require measurement.

616 For a smooth crossing at nonzero speed, evaluating the stationary branch  
617 along the changing control gives a deterministic tracking approximation that  
618 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

619 This is a deterministic tracking approximation from Eq. (3) at fixed  $D$ ; the  
620 crossing controls in Section 8.2 show that it fails as an observed-onset ap-  
621 proximation under the tested initialization. A separation of geometry and  
622 phase time scales can support tracking away from threshold. The awakening  
623 hypothesis needs a calibrated dynamical prediction for an operational onset  
624 above the finite-size fluctuation floor. Crossing  $K_c$  alone does not supply a  
625 recovery latency.

### 626 **9.3 A protocol for the awakening experiment**

627 Neural inertia describes a behavioural barrier and hysteresis between anaes-  
628 thetic induction and emergence (Friedman et al., 2010); recovery can pass  
629 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,  
630 2018). The  $v^{1/2}$  exponent of an ideal delayed bifurcation is generic (Baer  
631 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field  
632 mechanism from a synaptic one. A possible discriminator is spatial onset  
633 relative to independently estimated local field coupling and network con-

nectivity. The field-based onset map below is a separate model from the mean-field one, specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions  $i$  and measure pre-onset order  $r_i(t_0)$  and diffusion  $D_i$ . Define a field score  $h_i = N_i \mathcal{T}_i E_i f_i$  from independent pre-onset field and timing measurements, and a connectivity score  $c_i$  from a separately estimated structural network. For the same measured emergence drive  $g(t)$ , compare  $K_i^F(t) = a_F g(t) h_i$  with  $K_i^C(t) = a_C g(t) c_i$ . Each model has one fitted scale. The local first-harmonic approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

Predicted onset is the first crossing of a region-specific threshold fixed from pre-onset fluctuations. Fit the two scales on training sessions, then compare held-out onset times and site order under the same observation model. Larger field scores predict earlier onset only when baseline order and diffusion do not reverse the ranking. This declared approximation omits interregional coupling and saturation; a successful ranking would support its field covariate, not prove ephaptic causation. Supplemental Material specifies the controls and failure conditions.

A within-subject protocol can manipulate emergence rate, with spatially resolved recordings to compare the proposed field and connectivity models:

1. **Manipulate emergence rate.** Vary the propofol infusion taper

654 across sessions within the same subject, recording drug concentrations  
655 or a validated pharmacokinetic proxy to quantify the manipulation.

656 2. **Measure behavioural recovery.** Record time-to-response to a stan-  
657 dard stimulus as the independent behavioural endpoint, rather than  
658 estimating  $\Delta K$  directly.

659 3. **Resolve spatial onset.** Identify anatomical locations where coher-  
660 ence first crosses the baseline-scaled threshold and score the two maps  
661 of Eq. (12) on held-out sessions. The two covariate maps are fixed  
662 before the outcome sessions are inspected.

663 4. **Match physical controls.** Confirm that the observed coherence-to-  
664 amplitude relation holds under the specific local diffusion  $D$  and ob-  
665 servation noise. The model’s quadratic-variation rate  $2D$  applies to  
666 the latent phase; a band-limited analytic phase is smooth below its  
667 correlation time, and on a constructed field its squared-increment rate  
668 changes 37-fold between 0.2-ms and 20-ms steps. Estimating  $D$  there-  
669 fore requires an explicit observation model for filtering and sampling.

670 Predeclare the recording window, regional fluctuation thresholds, and a  
671 shared timing-error model. Fit the two scales and that observation model  
672 on training sessions only; on held-out sessions, score the observed site-onset  
673 times and sites that never cross by the same regional predictive rule. Report  
674 site order as a secondary outcome and behavioural time-to-response sepa-  
675 rately. Sessions without a crossing are censored observations, not zero delays

676 or exclusions. Compare the two scores only where independently estimated  
677 field and connectivity maps predict distinguishable onset patterns at the  
678 recording resolution. A held-out field-model advantage would support that  
679 specified spatial predictor; it would not identify an ephaptic cause, shared  
680 content or experience. Supplemental Material gives the controls and pilot  
681 quantities needed to judge whether the comparison is informative.

## 682 **10 Discussion**

### 683 **10.1 Unity and a minimal self as interpretive motiva-** 684 **tion**

685 We propose the state glued from compatible local descriptions as a formal  
686 correlate of conscious unity, and its accurate internal reconstruction as a cor-  
687 relate of a minimal self. In the cup example, this would mean one description  
688 that preserves the cup's relation to the perceiver as well as the scene. Com-  
689 patibility permits distinct contents and uncertainty; the shared quantities  
690 must be chosen from independent evidence about what each region encodes.  
691 The reconstruction need not be a verbal or autobiographical account.

692 The minimal self, in Gallagher's sense, is the pre-reflective sense of be-  
693 ing the subject of an experience, present before any narrative or reflective  
694 self-description (Gallagher, 2000). Two of its features motivate the proposed  
695 correlate. It is not a separate object of experience but a perspective built

696 into the experienced scene; correspondingly, the reconstruction is not a sec-  
 697 ond representation beside the glued state but a map whose fixed point is  
 698 that state, read through a part of it. It is immediate rather than inferred  
 699 from a higher-order state; correspondingly, the fixed point needs no further  
 700 state that represents the first. The correspondence is structural. Self-model  
 701 theory locates the phenomenal self in a transparent model a system has of  
 702 itself (Metzinger, 2003), and interoceptive-inference accounts ground it in  
 703 prediction of the body’s own states (Seth, 2013). The present condition is  
 704 narrower than either: it asks only that a declared local encoding recover the  
 705 global state, including perceiver-relative variables, at a declared fidelity, and  
 706 it says nothing about transparency or about which bodily variables must be  
 707 modelled. Either account could supply the family of states and the fidelity  
 708 that the condition leaves as inputs. Under E89 that family is one state at  
 709 zero error and a family of diameter at most  $2\varepsilon/(1 - \Lambda)$  within error  $\varepsilon$  (Sec-  
 710 tion 5.2), so under E89 the condition constrains a single self-state; a self that  
 711 persists through change needs the input-conditioned map of Section 5.2. Its  
 712 proposed relation to experienced perspective remains an interpretation. Nei-  
 713 ther identification is among the connecting hypotheses of the composition  
 714 (Table 1), and satisfying them would not measure experience. The unity  
 715 proposal is also only as definite as its cover. The gluing theorem holds for  
 716 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover  
 717 of interest overlaps in content rather than in space. Section 4 proposes fixing  
 718 that cover by decodability on held-out data, which excludes covers chosen for

719 agreement and makes non-degeneracy a testable property of the recordings.  
720 Its variables, criterion and decoder class are pre-registered by rules that never  
721 consult agreement.

## 722 **10.2 A comparison that could disconfirm the link to** 723 **content**

724 The framework predicts that conscious content tracks compatibility, not co-  
725 herence. The comparison must also separate compatibility from how well  
726 each region encodes the stimulus, because two accurate decoders agree auto-  
727 matically. If decoders  $d_A$  and  $d_B$  estimate a shared stimulus quantity  $x$  with  
728 errors  $e_A = d_A - x$  and  $e_B = d_B - x$ , then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

729 Consciously perceived stimuli are usually decoded better, so higher agreement  
730 on those trials would follow from single-region fidelity alone. At matched  
731 single-region error, agreement differs only through the covariance of the two  
732 errors. The comparison adds one premise to link compatibility to that term:  
733 two regions describing one content err together, so that compatibility appears  
734 as error covariance beyond what single-region fidelity implies.

735 Present matched stimuli that are perceived on some trials and not on  
736 others, for example at masking or rivalry threshold, and fit decoders for one  
737 shared stimulus quantity independently on two regions that both encode it,

738 selected by the decodability criterion of Section 4 on separate data. Clas-  
739 sify trials by perceptual state with a no-report measure, such as reflexive  
740 eye movements or pupil responses that track the dominant percept in rivalry  
741 (Tsuchiya et al., 2015), so that the classification does not also select for the  
742 neural correlates of reporting; report-based classification is a secondary anal-  
743 ysis. Match the two classes on phase coherence between the regions and  
744 on each region’s own decoding error (or, where the stimulus value is uncer-  
745 tain, on each region’s signal strength). The statistic is the excess agreement  
746 over a fidelity-matched null: pair region  $A$ ’s decode on one trial with region  
747  $B$ ’s decode on another trial of the same stimulus and error stratum, which  
748 preserves both marginal accuracies and removes any coupling of the errors.

749 The proposed link predicts a larger excess on perceived trials at matched  
750 coherence and matched single-region accuracy (Fig. 5). Equal excess across  
751 classes, or excess that follows coherence and not perceptual state, counts  
752 against it. A positive result is not yet specific: noise shared upstream of both  
753 regions also correlates their errors, so the excess should be compared against  
754 an upstream region’s decoding error, matched in the same way. A self-model  
755 comparison has the same form: at matched accuracy for the scene variable,  
756 the reconstruction of perceiver-relative variables, such as hand position rel-  
757 ative to the target, from a declared local encoding should be more accurate  
758 on perceived trials. These are correlational tests of proposed correlates, and  
759 the no-report classification is itself an indirect marker of experience.

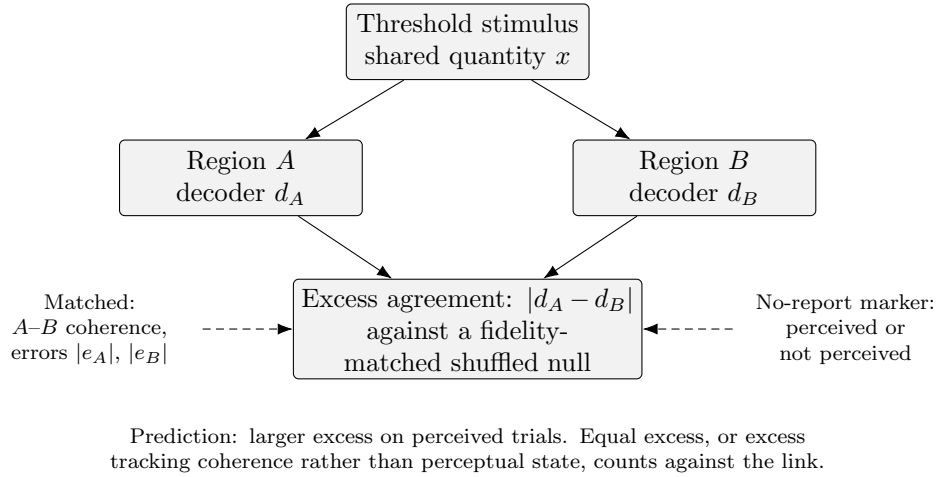


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

### 10.3 What a test of the full proposal must establish

A test needs separate estimates of effective coupling and diffusion, shared content on overlapping regions, and reconstruction from a specified encoding at a declared fidelity. The cover, relevant state family and decoder must be fixed without selecting them for agreement. An independent experiential endpoint is needed to evaluate the interpretation; absence of a report alone does not define absence of experience. A unified episode with incompatible decoded descriptions would challenge the unity identification. A unified episode lacking the proposed self-encoding would challenge the minimal-self claim. Conversely, satisfying the mathematical conditions without such an

770 episode would challenge their sufficiency as correlates.

771 For the physical route, cortical actuator modes and prices, the geometry–  
772 coupling relation and a dynamical onset model remain to be measured. The  
773 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-  
774 diction. Controlled changes in coordination and independently decoded con-  
775 tent agreement would address the representational connection. The synthetic  
776 decoder exercise calibrates part of these measurements, and the EEG exercise  
777 identifies the estimand a spatial test needs.

778 Integrated-information, global-workspace and predictive-processing ac-  
779 counts organize unity, access and modelling differently (Tononi et al., 2016;  
780 Baars, 1997; Dehaene et al., 2014; Friston, 2010). This account specifies con-  
781 ditional relations among coupling, local content and reconstruction, each tied  
782 to a measurement.

## 783 11 Limitations

784 *Experience and interpretation.* Synchrony is not itself an experiential mea-  
785 sure, and nothing here establishes a consciousness criterion: the conditions  
786 are proposed correlates to be tested against report or another independent  
787 endpoint. The mathematical states are spatial measures and fields, and iden-  
788 tifying them with perceptual content requires independent decoding evidence.  
789 The constructions of Section 4 select neither the neural cover nor the decoder,  
790 and identifying their contents with perceptual content, or establishing that

791 the identification bears on experience, is a further modelling task. The cor-  
792 respondence between the reconstruction’s fixed point and the minimal self is  
793 structural and does not show that a fixed point is experienced as a perspec-  
794 tive. The remaining explanatory question is whether these relations occur in  
795 neural activity and matter for experience; the framework does not explain  
796 why satisfying them would be accompanied by experience.

797 *Formal scope.* The formal results are conditional on declared encoders,  
798 covers and maps whose cortical counterparts are unmeasured. Each edge  
799 of the composition is supplied rather than derived from the one above it,  
800 so the composition does not establish that self-representation follows from  
801 coherence. The joint witnesses satisfy every edge in a specified model: they  
802 prove neither that cortex satisfies those hypotheses nor that the glued state is  
803 experienced, and establish no common cortical mechanism. The contraction  
804 factor  $\Lambda$  is stipulated, not derived. It borrows the rate  $(K - 2D)/2$  at which  
805 the first harmonic of the uniform density grows (Section 3), a rate about a  
806 different state from the one  $F$  acts on, and no result derives a Lipschitz law  
807 for an encoding or readout from the phase dynamics; constructing a map with  
808 that factor remains a modelling obligation. No physical input-conditioned  
809 map  $F_u$  is constructed, the gain  $L$  of Section 5.2 is derived here for no physical  
810 readout, and E89 supplies no resolution of a declared family by the encoding  
811 region. Agreement of a common projection does not establish agreement of  
812 the full vector, and the general vector result of Section 4 supplies no identi-  
813 fication with the spatial-measure sheaf. The cover on which compatibility is

814 assessed is fixed only relative to pre-registered variables, decoder class and  
815 criterion, and the phase-to-content bounds are stated for spatial covers rather  
816 than for the content cover that criterion yields. No result connects exact or  
817 approximate compatibility to correlated decoding errors; the comparison of  
818 Section 10.2 takes that link as a premise.

819 *Physical identification.* The formal field condition requires a continuous  
820 kernel, a normalized substrate, a specified average coupling and noise, and a  
821 site-contribution bound. Those conditions do not identify an electromagnetic  
822 substrate; the extracellular interpretation needs the separate calibration of  
823 Section 9, and the field commitment requires physical evidence beyond a  
824 choice of mathematical representation. Phase organization under propofol  
825 does not measure coupling, and the support comparisons in Supplemental  
826 Material concern specified support constraints and supply neither that evi-  
827 dence nor energy prices. Neither the conversion rate  $\gamma$  nor the installation  
828 factor  $\kappa$  follows from a heat bound, and no cortical value of  $\kappa$  is known.  
829 Testing the physical threshold requires independent estimates of the inter-  
830 action kernel, its aggregation and diffusion in the same cortical regime. No  
831 measured connectivity yet supplies the spectral route of Section 4.2. The  
832 mean-field model has no spatially varying  $\kappa$  and derives no nucleation or-  
833 der, which is why the onset comparison of Section 9.3 rests on a separately  
834 specified model.

835 *Numerical and empirical controls.* The controls of Section 8 are finite  
836 regimes on declared models and calibrate no cortical quantity. A slow con-

837 trol variable supplies no general guarantee of branch tracking; the winding  
838 sweep establishes no transition and no biological identification; phase agree-  
839 ment cannot replace an independently decoded content measurement. The  
840 synthetic decoder exercise validates a measurement on known constructions,  
841 and the EEG exercise checks a measurement pipeline: it identifies what a  
842 pooled bipolar estimand cannot measure, and neither exercise identifies a  
843 neural content code, tests experience or validates any experiential claim.

## 844 12 Conclusions

845 Phase coherence, compatibility and self-reconstruction are separable condi-  
846 tions. Coherence constrains shared content only through encoders shared on  
847 overlaps and with a bound that grows as the square root of the population  
848 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch  
849 order; chaining it through overlapping patches reaches 0.312 mm of that  
850 sheet, and a spectral route removes the population count only for a declared  
851 connectivity at a frequency-locked configuration. Across the centimetre sep-  
852 arations of distinct cortical territories, the positive link from coherence to  
853 content is the bridge assumption E78 rather than a derived result. Com-  
854 patible local descriptions, not synchrony, determine one global state; and a  
855 contracting self-map fixes that state while reconstructing only a family of  
856 bounded diameter. Each result replaces an assumption about synchrony by  
857 a measurement: independently decoded agreement on overlaps, a declared

858 encoding and its fidelity, and calibrated coupling and diffusion. The compar-  
859 ison of Section 10.2 states how such measurements could count against the  
860 proposed link to conscious content.

## 861 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

| Symbol                    | Defined        | Meaning                                                                                                                 |
|---------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------|
| $\theta_i, \omega_i$      | Eq. (1)        | Phase and natural frequency of oscillator $i$                                                                           |
| $N$                       | Eq. (1)        | Number of oscillators in the population the quantity is read on                                                         |
| $D$                       | Eq. (1), Supp. | Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$      |
| $K_{ij}, K$               | Eq. (1)        | Pairwise and scalar mean-field coupling, $K_{ij} = K/N$                                                                 |
| $K_c$                     | Eq. (2)        | Synchronization threshold, $K_c = 2D$                                                                                   |
| $r, r_N$                  | Eq. (1)        | Resultant length (order parameter), on the population it is read on                                                     |
| $a, \psi$                 | Eq. (2)        | Concentration $a = Kr/D$ and collective phase of the stationary density                                                 |
| $q$                       | Section 8.3    | Winding number of a phase field read around a ring                                                                      |
| $U_{\text{inst}}, \kappa$ | Section 3.2    | Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by $\kappa U_{\text{inst}}$ |
| $L$                       | Section 4.2    | Lipschitz constant of an encoder or readout                                                                             |
| $\varepsilon, \delta$     | Section 4.2    | Content and observation discrepancy tolerances                                                                          |
| $M, b$                    | Eq. (9)        | Size of a declared family of states and code length in bits                                                             |

Table 2 (continued)

| Symbol                   | Defined               | Meaning                                                                                                                  |
|--------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\Lambda, \tau$          | Eq. (7)               | Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), $\tau$ is the duration of a change |
| $J, \sigma_J, C$         | Eq. (4)               | Probability current of the phase density, its current functional and a current budget                                    |
| $T$                      | Section 5             | Number of communication rounds to a deadline                                                                             |
| $\nu$                    | Supplemental Material | Fan-in of a communication graph: the most sites any one site hears from                                                  |
| $\gamma, \mathcal{T}, f$ | Eq. (10)              | Field-to-coupling conversion rate, timing sensitivity and frequency                                                      |
| $\alpha$                 | Section 9             | Extracellular volume fraction                                                                                            |
| $\lambda$                | Section 9             | Extracellular tortuosity                                                                                                 |
| $\Omega, \lambda_2$      | Section 4.2           | Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph           |
| $\beta, k$               | Section 4.2           | Chord diameter of a patch and the diameter of a cover's nerve, in hops                                                   |
| $\mathcal{C}, e_{i,x}$   | Section 4.2           | Content space and the phase encoder patch $i$ reads at site $x$                                                          |
| $\Phi, R$                | Eq. (9)               | Encoder and readout of the reconstruction criterion                                                                      |
| $\mathcal{A}, F$         | Eq. (7)               | Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$                                     |
| $E$                      | Eq. (10)              | Field amplitude, in the cortical identification                                                                          |
| $s, s_*$                 | §4–5                  | A global section and the one the reconstruction fixes                                                                    |

## 862 Data and code availability

863 The Lean development, simulation scripts, tests and saved numerical sum-  
864 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`  
865 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-

lib commit 5ce203ecd53f8c3e47ee5530d7c86b94a55f0183. Publication  
simulation values are generated from saved summaries without rerunning  
production sweeps. The exploratory EEG analysis uses OpenNeuro dataset  
ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,  
1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing  
and selection are described in Supplemental Material and implemented in  
simulations/empirical\_collapse.py. The repository contains the repro-  
ducible analysis and does not redistribute the raw recordings.

## **Funding and competing interests**

This research received no external funding. The author declares no compet-  
ing interests.

## **Use of AI tools**

AI assistance to this work was substantial and is not separable by task.  
Several coding assistants and chat interfaces were used in an interleaved way  
throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,  
Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-  
5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),  
Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek  
v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them

885 they contributed to the Lean development, the simulation and analysis code,  
886 and the drafting and revision of both documents. Because the tools were  
887 used in combination and in alternation, no section, proof, figure or result is  
888 attributable to any one of them, and none is claimed to be. The author is  
889 responsible for all content, including everything a tool produced. Machine  
890 checking establishes the stated Lean results under their hypotheses; it does  
891 not validate biological interpretations or replace scientific review of generated  
892 code and prose.

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